

Four-Core Pipelined Implied Volatility & Greeks Engine (Artix-7 200T @ 100 MHz)

AXI4-Stream Ingress  
(256-bit S, K, T, r, C\_mkt, TID)

Stage 1: Ingress Registration  
& Queuing (1 Cycle)

- AXI4-Stream 256-bit Ingress: {S, K, T, r, C\_mkt, TID}
- Single-Cycle Register + TID Tag
  - No Ingress Divider (Zero Division Latency)
- Q8.24 Fixed-Point Precision

Contract

Stage 2: Analytical Seeder  
(Brenner-Subrahmanyam, 64 Cyc)

$$\sigma_0 \approx \frac{2,5066 C_{mkt}}{\sqrt{T} (S + K)/2}$$

- Shared Digit-Recurrence  $\sqrt{T}$  (29 Cyc, II = 1)
  - Forwards  $\sqrt{T}$  Directly to Stage 3
- Non-Restoring Divider (33 Cyc, II = 1)
- Analytical Initial Estimate (Sec 2.4):

Med Abs Err: 0.1037 vol (1,037 vol-bps) | Rel: 30.52% (11.53% ATTC)

$\sigma_0, \sqrt{T}$  (Forward)

Stage 3: Core Feedforward Datapath  
(126 Cycles, II = 1)

- II = 1 Pipeline (Reused Across Passes)
- Scale-Invariant Padé Ratio (S-K)/(S+K)
- 18-Cyc CORDIC Log Fallback (SRL15-Matched)
  - Horner 5th-Order Poly CDF N(d1), N(d2)
- Hyperbolic CORDIC for  $\exp(-rT)$  &  $\exp(-d1^2/2)$

Stage 4: NR Update & Dividers  
(33 Cycles, II = 1)

- Convergence:  $|C_{BS} - C_{mkt}| \leq \$0.01$
- **u\_nr\_divider**:  $\Delta\sigma = (C_{BS} - C_{mkt})/\nu$
- **u\_gamma\_divider**:  $\Gamma = \phi(d_1)/(S\sigma\sqrt{T})$
- Greeks:  $\Delta = N(d_1)$ ,  $\nu = S\sqrt{T}\phi(d_1)$
- TID-Scoreboard Priority Loopback

Converged → Egress

Nonconverged →  $\sigma_{n+1}$  Recirculation  
(Priority Restore to Stage 3)

$C_{BS}, \nu$   
(126 Cyc)

Stage 5: Multi-Core Arbiter  
& 128-Bit Egress (2 Cycles)

- Round-Robin 4-Core Drain Arbiter
  - 128-Bit Data Payload:
    - [127:122] Local TID (6b)
    - [121:96] Delta Greek (26b, Q2.24)
    - [95:64] Gamma Greek (32b, Q8.24)\*
    - [63:32] Vega Greek (32b, Q8.24)\*
    - [31:0] sigma Implied Vol (32b, Q8.24)
  - Sideband: Core ID (2b)
- \*Carries  $\nu^*, \Gamma^*$  for normalized inputs

AXI4-Stream Egress  
(128b Payload + 2b Core ID Sideband)

Zero-BRAM Context Store

- 64 Context Slots / Core (256 Total)
- 60 Active + 4 Headroom Invariant (Prevents Recirculation Deadlock)
  - Distributed LUTRAM & SRL32
  - 0 Block RAM / UltraRAM in Core
- Leaves 365 BRAM36 Free for System

Context